



TASK 4: Analytics & Tool Match

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Section: 6A CSE (DS)

Course: Big Data Analytics

Smart City Analytics & Tool Match Table

Business Question	Analytics Type	Tools
What happened to city traffic yesterday?	Descriptive Analytics	BI Tools, Hadoop
Why did water consumption increase in a zone?	Diagnostic Analytics	Spark, Hadoop
What will be the power demand next month?	Predictive Analytics	Spark (MLlib), Hadoop
What action should be taken to reduce traffic congestion?	Prescriptive Analytics	Spark + NoSQL

Description for the Table:

This table explains how analytics supports decision-making in a Smart City environment. It connects common city management questions with the appropriate type of analytics and the technologies used to perform them. The process begins with understanding past events, then identifying causes, predicting future outcomes, and finally recommending suitable actions.

Descriptive analytics summarizes past city data such as traffic or energy usage. Diagnostic analytics examines the reasons behind unusual patterns or sudden changes. Predictive analytics applies models to estimate future trends like power demand or traffic flow. Prescriptive analytics goes further by suggesting the best actions, such as traffic rerouting or energy optimization strategies.